PM2.5 study: findings summary

County comparisons, weather relationships and next-day prediction
Alabama county-labeled monitoring series | January-December 2025

Data: 10 worksheets, 365 dates per county and 3,591 observed PM2.5 values. There are 59 missing and 11 negative readings. Negative readings were retained in the main analysis, with explicit sensitivity checks. **All values use unverified workbook units.**

1. County comparisons

County	Mean	Median	95th percentile	Maximum
Jefferson	10.12	8.80	20.80	39.90
Etowah	8.19	7.60	14.53	25.80
Montgomery	8.07	7.60	13.72	30.20
Mobile	7.34	7.20	11.89	21.60
Tuscaloosa	7.03	6.60	13.10	25.20
Sumter	7.03	6.80	12.18	16.40
Morgan	6.87	6.35	12.53	18.50
Clay	6.71	5.90	13.36	48.50
Baldwin	6.62	6.70	11.43	24.50
DeKalb	6.30	5.80	12.10	20.20

Jefferson had the highest mean and median; its mean was approximately 61% higher than DeKalb's. Clay had the highest single-day value (48.50) despite a comparatively low mean (6.71), indicating that typical conditions and extreme episodes require separate interpretation.

Exploratory 95% intervals for the mean were 9.24-10.97 in Jefferson and 5.80-6.83 in DeKalb. Many middle-ranked counties had overlapping intervals. Intervals used 1,000 resamples of seven-day calendar blocks; paired-date comparisons and block-length sensitivity are available. Excluding negative readings changed means only slightly.

2. Weather relationships

Wind was the most consistent negative association. In models adjusting for month and other weather variables, wind coefficients were negative with multiple-comparison-adjusted $q < 0.05$ in 9/10 counties. A one-standard-deviation wind increase corresponded to estimated PM2.5 differences of -1.47 in Montgomery, -1.41 in Jefferson and -1.39 in Etowah.

Warmer days within a month tended to have higher concentrations in several counties. Monthly-demeaned temperature correlations were 0.45 in DeKalb and approximately 0.40 in Morgan and Tuscaloosa. Temperature, dew point and humidity were strongly redundant: large individual coefficients are unstable and should not be interpreted independently.

Wind coefficients stayed negative across the reported sensitivity scenarios, although statistical support varied. Rainfall evidence was less consistent; DeKalb had a negative adjusted association, and Sumter's constant rainfall column could not support an estimate. Baldwin and Mobile shared identical weather values on jointly observed dates.

3. Prediction results

Forecast question: use observations through yesterday to predict today's PM2.5.

Evaluation: expanding monthly validation in July-September selected model settings and the model family for each county. Models were then fitted on January-September and evaluated on October-December, without using test data to fit preprocessing or select models. Evaluation covered 897 county-days shared by all approaches.

Inputs were PM2.5 lags of 1, 2 and 7 days, previous-day weather and known calendar features. Models remained fixed during the test period, while lagged observations updated daily. This is one-day-ahead evaluation, not a three-month forecast from a single starting date.

Held-out accuracy

Approach	MAE	RMSE
Training-period mean	2.878	3.498
Persistence: yesterday's PM2.5	2.476	3.299
Ridge: PM2.5 lags + calendar	2.507	3.104
Ridge: adding lagged weather	2.281	2.868
Ridge: weather PCA components	2.298	2.867
Random forest	2.270	2.876
Validation-selected model per county	2.276	2.876

MAE is mean absolute error: the average distance between prediction and observation. **RMSE** gives greater weight to large errors. Lower is better; both use workbook concentration units.

What the comparison supports

Modest overall improvement: the validation-selected models reduced pooled MAE from 2.476 to 2.276, approximately 8%. They improved on persistence in 7/10 counties, but performed worse in Baldwin, Mobile and Sumter.

Weather added predictive information in the ridge comparison: MAE fell from 2.507 with PM2.5 lags/calendar to 2.281 after adding lagged weather. Weather PCA did not improve pooled accuracy over the original weather variables (2.298 versus 2.281), although it helped some counties.

County improvements remain uncertain: paired seven-day bootstrap intervals were entirely below zero in Montgomery and Morgan, and only just below zero in Etowah. Other apparent gains were not clearly separated from temporal variability. These are pointwise, exploratory intervals.

Systematic overprediction remains: the selected models had pooled prediction-minus-observation bias of +0.87, reaching +2.22 in Jefferson. Lower MAE does not imply unbiased forecasts. Models that looked best on the test months were not always those selected beforehand; changing selection now requires a new independent test.

Interpretation limits and source

These observational, one-year results describe the supplied monitoring series. They do not establish causal weather effects, identify pollution sources, represent exposure across entire counties, demonstrate recurring seasonality or long-term trends, or support regulatory/health classifications. Units, station provenance and reporting delays need verification. Another year is needed for stronger external forecast validation.

Source: Meterological_data_ALAbama_copy.xlsx; analysis folders 01_county_comparison, 04_weather_associations and 07_prediction under PCA_SR/initial_study. Detailed methods, quality flags, uncertainty and saved outputs accompany this summary. The analysis passed 99 numerical/structural checks; figures were visually reviewed.